

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 9-2026: A 12-Year-Old Girl with Altered Mental Status and Hypoglycemia

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PRESENTATION OF CASE

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CME



Dr. Maya Yogev Lifshitz (Pediatrics): A 12-year-old girl was admitted to the pediatric service of this hospital because of altered mental status, abdominal distention, and persistent hypoglycemia.

On the morning of the hospital admission, the patient's mother had difficulty rousing the patient. On awakening, the patient could not maintain her gaze and was speaking nonsensically. Her grandmother reported that she had been well the night before, when she was noted to have eaten a full meal and was behaving normally. The patient was transported by emergency medical services to the emergency department of this hospital.

On presentation to the emergency department, the patient had a temporal temperature of 35.4°C; the vital signs were otherwise normal. She appeared ill and was intermittently responsive, at times answering questions with one-word responses. There were intermittent periods of stupor. Spontaneous movement of the arms and legs was noted. The blood glucose level was 30 mg per deciliter (1.7 mmol per liter; reference range, 70 to 110 mg per deciliter [3.9 to 6.1 mmol per liter]); other laboratory test results are shown in Table 1. Treatment with an ampule of intravenous glucose in water was administered, with subsequent improvement in mental status, and a continuous intravenous infusion of dextrose in water was started.

Dr. Pallavi Sagar: Computed tomography (CT) of the head, performed without the administration of intravenous contrast material, showed no hydrocephalus, intracranial mass, or parenchymal hemorrhage.

Dr. Yogev Lifshitz: On admission to the hospital, the patient reported no ongoing symptoms or previous similar episodes and no ingestion of any recreational drugs. Approximately 1 year before this admission, she had had unintentional weight loss of 4 kg during a period of intense running; the weight was later partially regained. More recently, the patient's mother had observed that the patient had had early satiety, noting that she often left meals unfinished. A review of systems was negative for constipation, fever, dizziness with standing, cough, coryza, diarrhea, night sweats, changes in skin pigmentation, rigors, nausea, and vomiting. Medical history was notable for vestibular migraines that had begun at 7 years of age. The patient took no medications, and immunizations were up to date. She had no known adverse reactions to medica-

tions. She lived at home in suburban Massachusetts with her mother, sister, and grandmother and had no pets. She was on the track team at middle school and was starting seventh grade without academic or developmental issues. Menarche had occurred earlier that year. Family history was notable for prostate cancer, breast cancer, and hypertension in her maternal grandparents.

On admission to the pediatrics service, the patient had an oral temperature of 37.0°C, a blood pressure of 121/61 mm Hg, a heart rate of 89 beats per minute, a respiratory rate of 19 breaths per minute, and an oxygen saturation of 97% while she was breathing ambient air. The weight was 43 kg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) was 18.7. The mucous membranes were moist, and no lesions were detected in the oropharynx. No murmur was noted on cardiac auscultation, and the lungs were clear. The abdomen was distended, with no tenderness on palpation; a masslike fullness was present in the pelvis. There was no rash or palpable lymphadenopathy. The patient was alert and oriented with a normal gait and no motor or sensory deficits.

Subsequent measurement of the blood glucose level showed persistent hypoglycemia, with levels ranging from 50 to 65 mg per deciliter (2.8 to 3.6 mmol per liter), despite the continuous intravenous infusion of dextrose. During a period of hypoglycemia, the levels of β -hydroxybutyrate, insulin, and C-peptide were below the limits of detection and the proinsulin level was 0.6 pmol per liter (reference range, 3.6 to 22.0). Additional laboratory test results are shown in Table 1. Blood specimens were obtained for culture. Urinalysis was negative for ketones, cells, and protein.

Diagnostic tests were performed, and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Katherine Lord: This 12-year-old girl meets the criteria for Whipple's triad for the diagnosis of hypoglycemia: she presents with signs and symptoms of hypoglycemia, including altered mental status and hypothermia; she has a corresponding low blood glucose level of 30 mg per deciliter; and her symptoms resolve with normalization of the blood glucose level.¹ The differential diagnosis of hypoglycemia in childhood is broad. To narrow the differential diagnosis, levels of hormones and metabolic fuels are measured at the time of hypo-

glycemia. Testing of blood and urine obtained during an episode of hypoglycemia, known as the "critical sample," can guide additional testing. The patient has a normal lactate level and liver-function test results that are not suggestive of disorders involving gluconeogenesis and glycolysis. The strong cortisol response to cosyntropin administration rules out adrenal insufficiency.

The suppressed β -hydroxybutyrate level seen in the critical sample points toward hyperinsulinemia or insulin-mediated hypoglycemia as the cause of the patient's hypoglycemia. Supporting this diagnostic category is her increased glucose use, as evidenced by the presence of hypoglycemia despite receipt of an intravenous dextrose infusion. As I will discuss below, the patient's undetectable insulin and C-peptide levels do not rule out specific causes of inappropriate insulin secretion or signaling. Another important clinical clue is the patient's age, which makes an acquired form of hypoglycemia more likely than a congenital one.

INSULINOMA

When insulin-mediated hypoglycemia manifests in later childhood or adolescence, an insulinoma is high on the differential diagnosis. Insulinomas are insulin-producing tumors of the pancreatic beta cells and often cause fasting hypoglycemia. A majority of insulinomas are benign, and they can occur in association with multiple endocrine neoplasia type 1. Because insulinomas are small tumors, they can be challenging to detect in the pancreas; multiple imaging studies are often used to localize these tumors.² Given their small size and the fact that they are located in the pancreas, they rarely cause abdominal distention, which is described in this patient. In addition, if the patient had an insulinoma, I would expect her to have detectable insulin and C-peptide levels at the time of hypoglycemia, as well as an elevated proinsulin level³; therefore, insulinoma is an unlikely diagnosis in this patient.

FACTITIOUS HYPOGLYCEMIA

It is important to consider surreptitious administration of insulin or sulfonylureas in children (after infancy) and adolescents presenting with hypoglycemia. Parents, caregivers, or the child may be administering the medications to induce hypoglycemia. The onset of the hypoglycemia in such cases is abrupt. However, with surreptitious administration of these agents, the pattern of hypoglycemia is typically erratic, waxing and waning

Table 1. Laboratory Data.*

Variable	Reference Range, Children	On Initial Presentation
Blood		
Hemoglobin (g/dl)	11.5–15.5	14.2
Hematocrit (%)	35.0–45.0	40.5
Mean corpuscular volume (fl)	77.0–95.0	76.4
White-cell count (per μ l)	5000–14,500	6040
Differential count (per μ l)		
Neutrophils	1500–8000	4730
Lymphocytes	1500–7000	2140
Monocytes	200–1600	260
Eosinophils	0–1200	10
Basophils	0–400	10
Platelet count (per μ l)	150,000–450,000	280,000
Sodium (mmol/liter)	135–145	140
Potassium (mmol/liter)	3.4–5.0	3.8
Chloride (mmol/liter)	98–108	105
Carbon dioxide (mmol/liter)	23–32	23
Urea nitrogen (mg/dl)	5–20	4
Creatinine (mg/dl)	0.30–1.00	0.37
Glucose (mg/dl)	70–110	30
Calcium (mg/dl)	8.5–10.5	9.2
Aspartate aminotransferase (U/liter)	10–40	37
Alanine aminotransferase (U/liter)	10–50	22
Alkaline phosphatase (U/liter)	129–417	137
Total bilirubin (mg/dl)	0.0–1.0	0.3
Albumin (g/dl)	3.3–5.0	4.4
Globulin (g/dl)	1.9–4.1	3.5
Cortisol, morning (μ g/dl)	5.0–25.0	5.1
Cortisol, after cosyntropin administration (μ g/dl)	>15.0	20.6
Uric acid (mg/dl)	2.4–5.7	2.4
Insulin (μ U/ml)	2.6–25.0	<0.4
C-peptide (ng/ml)	1.1–4.4	<0.1
Proinsulin (pmol/liter)	3.6–22.0	0.6
β -Hydroxybutyrate (mmol/liter)	0.0–0.4	<0.1
Thyrotropin (μ U/ml)	0.55–4.78	0.64
Lactate (mmol/liter)	0.5–2.0	1.0
Urine		
Color	Yellow	Yellow
Clarity	Clear	Clear
pH	5.0–9.0	6.5
Specific gravity	1.001–1.035	1.009
Glucose	Negative	Negative
Ketones	Negative	Negative
Leukocyte esterase	Negative	Negative
Nitrite	Negative	Negative
Blood	Negative	Negative
Protein	Negative	Negative
Erythrocytes (per high-power field)	0–2	0

Table 1. (Continued.)

Variable	Reference Range, Children	On Initial Presentation
Leukocytes (per high-power field)	<10	0
Bacteria	None	None

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for cortisol to nanomoles per liter, multiply by 27.59. To convert the values for uric acid to micromoles per liter, multiply by 59.48. To convert the values for insulin to picomoles per liter, multiply by 7.175. To convert the values for lactate to milligrams per deciliter, divide by 0.1110.

in a nonphysiologic manner. The hallmark laboratory finding associated with exogenous insulin administration is a detectable insulin level without a detectable C-peptide level, which is not consistent with this patient's results.

In addition, both the insulin level and the C-peptide level are detectable with sulfonylurea administration. Routine drug screening tests do not detect sulfonylureas; a specialized blood test for sulfonylurea is required. However, this patient's undetectable insulin and C-peptide levels are not consistent with the administration of sulfonylurea or insulin.

CONGENITAL HYPERINSULINISM

Congenital hyperinsulinism is the most common cause of persistent hypoglycemia in children.⁴ Genetic defects in the beta cells result in dysregulated insulin secretion and severe hyperinsulinemic hypoglycemia. Congenital hyperinsulinism is typically diagnosed at birth or during infancy; however, some forms of the genetic disorder may occur in adolescence or adulthood.⁵ A common pitfall in the diagnosis of congenital hyperinsulinism is the expectation that insulin and C-peptide levels are detectable. Suppression of β -hydroxybutyrate and a positive glycemic response to glucagon therapy are more sensitive markers of congenital hyperinsulinism than blood levels of insulin and C-peptide.⁶ Hence, the patient's laboratory results do not rule out congenital hyperinsulinism as a possible cause of her hypoglycemia. However, given the patient's abrupt onset of symptoms, a congenital disorder is unlikely.

INSULIN AUTOIMMUNE SYNDROME

Insulin autoimmune syndrome is a rare disorder characterized by the development of autoantibodies against insulin.⁷ It occurs primarily in adults, although cases in children have been reported.⁷

Risk factors for insulin autoimmune syndrome include Japanese or Korean ancestry, a history of other autoimmune disorders, and exposure to certain classes of drugs. In patients with insulin autoimmune syndrome, insulin is secreted normally during a meal but is quickly bound by autoantibodies, which leads to transient hyperglycemia. The antibodies subsequently release the bound insulin in an unregulated manner, which results in hypoglycemia. With this underlying mechanism, hypoglycemia associated with insulin autoimmune syndrome occurs predominantly after meals. Fasting hypoglycemia, with which this patient presented, is less common. Patients with insulin autoimmune syndrome are reported to have very elevated insulin levels at the time of hypoglycemia. If insulin autoimmune syndrome is suspected, measurement of insulin autoantibodies is indicated. The absence of risk factors, the pattern of hypoglycemia, and the undetectable insulin level in this patient make this diagnosis unlikely.

NON-ISLET-CELL TUMOR HYPOGLYCEMIA

Non-islet-cell tumor hypoglycemia (NICTH) is a rare paraneoplastic process originating from solid tumors.⁸ Although this syndrome has been reported in adults, only a handful of cases in children have been reported in the literature. NICTH occurs most commonly in association with mesenchymal tumors of the chest and abdomen, although it has been described in patients with multiple types of solid tumors.⁹ The tumors are often large before NICTH develops, and they secrete high-molecular-weight insulin-like growth factor II (IGF-II), known as "big IGF-II."¹⁰ Big IGF-II is structurally similar to insulin and binds to the insulin receptor. Activation of the insulin-signaling cascade leads to increased peripheral glucose utilization, decreased hepatic glucose output, and ultimately, hypoglycemia.

In patients with NICTH, insulin and C-peptide levels are appropriately suppressed during hypoglycemia, as seen in this patient. They have a suppressed β -hydroxybutyrate level and a positive glycemic response to glucagon therapy owing to activation of the insulin-signaling pathway. The elevated IGF-II level suppresses growth hormone secretion, which leads to a low insulin-like growth factor I (IGF-I) level. However, the IGF-II level can sometimes be normal, depending on the sensitivity of the assay to detect the high-molecular-weight form. An elevated ratio of IGF-II to IGF-I is considered to be diagnostic of NICTH.¹¹ This patient's critical sample test results are highly suggestive of a diagnosis of NICTH, and her early satiety and fullness on abdominal examination suggest the possibility of a mass in her abdomen. As the next steps in the workup of this patient, I would perform abdominal imaging and obtain IGF-I and IGF-II levels to establish the diagnosis of NICTH.

DR. KATHERINE LORD'S DIAGNOSIS

Non-islet-cell tumor hypoglycemia.

DISCUSSION OF ENDOCRINOLOGIC MANAGEMENT

Dr. Rachel C. Whooten: The results of testing of the patient's blood sample obtained at the time of hypoglycemia were most consistent with a hyperinsulinemic state, characterized by the absence of ketones. This conclusion was supported by the results of a glucagon stimulation test, which showed that the blood glucose level had increased from 46 to 142 mg per deciliter (2.6 to 7.9 mmol per liter) after the administration of 1 mg of glucagon. During a state of insulin excess, administration of glucagon mobilizes glucose from stored glycogen. Thus, the brisk rise in the glucose level that was seen in this patient is consistent with insulin excess. However, given that she had very low or undetectable blood levels of insulin, C-peptide, and proinsulin, we suspected a process in which the biochemical profile mimics hyperinsulinism, such as that which occurs with NICTH. In this paraneoplastic process, excess production of IGF-II from a tumor results in an inappropriately high IGF-II level, decreased production of pituitary growth hormone, hypoglycemia due to increased glucose uptake in muscle and reduced glucose production in the liver, and decreased insulin production.¹¹⁻¹³

All these findings were identified in this pa-

tient, including an abnormally elevated ratio of IGF-II to IGF-I (206:24, or 8.6:1; reference value, <3:1). Imaging studies of the abdomen and pelvis were obtained to characterize the tumor that was considered to be the source of the NICTH. The patient received supportive care — including dextrose-containing fluids, glucose monitoring, and frequent meals containing a combination of fats, carbohydrates, and protein — while awaiting additional diagnostic evaluation to guide definitive treatment of the tumor.

IMAGING STUDIES

Dr. Sagar: CT of the abdomen and pelvis, performed after the administration of intravenous contrast material (Fig. 1), revealed a solid, enhancing intraperitoneal abdominopelvic mass that measured 28 cm by 21 cm, with cystic features but no calcifications or fat. The right ovary appeared normal; the left ovary was not well visualized. The liver, spleen, kidneys, adrenal glands, and pancreas appeared normal.

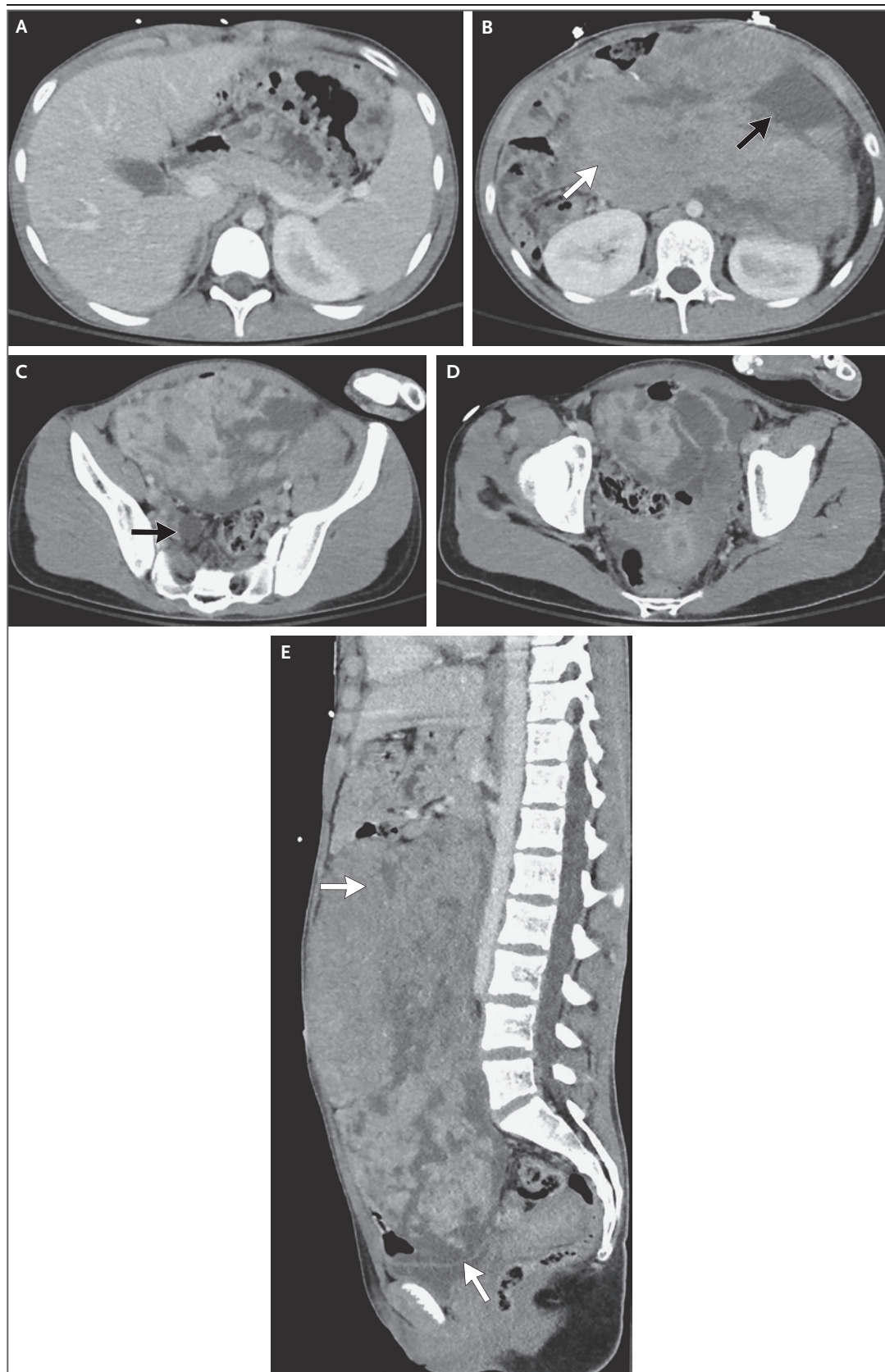
Contrast-enhanced magnetic resonance imaging (MRI) of the abdomen and pelvis (Fig. 2) was also performed, since it is superior to CT for soft-tissue characterization. MRI showed the solid, enhancing abdominopelvic mass with restricted diffusion, which was consistent with an aggressive hypercellular tumor. Normal ovaries were also noted. These characteristics appear to indicate the presence of a large extraovarian, intraperitoneal, solid abdominopelvic mass with aggressive features. On the basis of these findings, the differential diagnosis includes a mesenchymal tumor, carcinoma, metastatic sarcoma, lymphoma, and atypical teratoma.

DISCUSSION OF BIOPSY AND SURGICAL MANAGEMENT

Dr. Danielle B. Cameron: Cross-sectional imaging revealed a large, heterogeneous mass that appeared to emanate from the retroperitoneum, displacing

Figure 1 (facing page). CT of the Abdomen and Pelvis.

Shown are axial contrast-enhanced images of the upper abdomen (Panel A), midabdomen (Panel B), lower abdomen (Panel C), and pelvis (Panel D), as well as a sagittal reconstructed image (Panel E). A large enhancing abdominopelvic mass (Panels B and E, white arrows) with cystic components (Panel B, black arrow) is seen. The right ovary appears normal (Panel C, arrow).



the intestine and ovaries, and did not invade adjacent structures. In conjunction with the pediatric oncology team, the decision was made to proceed with a core biopsy rather than upfront surgical resection, given the size of the mass, the differential diagnosis, and potential systemic therapeutic options for certain causes of hypoglycemia on

the differential diagnosis. The goal of potential neoadjuvant therapy would be to shrink the mass and minimize the risk of complications associated with the operation. While awaiting the results of the core biopsy and molecular characterization, the patient continued to receive intravenous dextrose-containing fluids at night, and

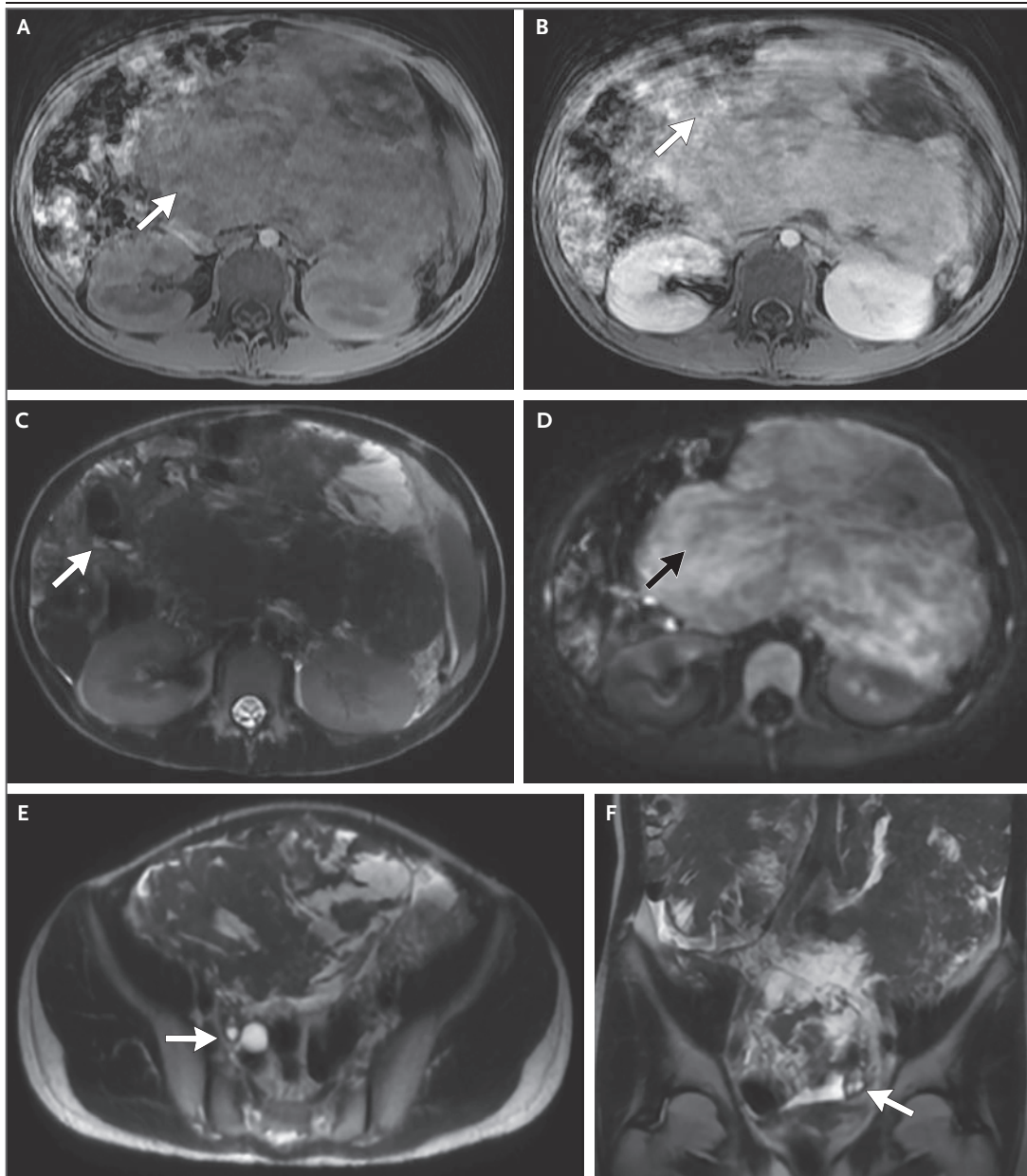


Figure 2. MRI of the Abdomen and Pelvis.

Axial T1-weighted images obtained before and after the administration of contrast material (Panels A and B, respectively), an axial T2-weighted image (Panel C), and an axial diffusion-weighted image (Panel D) show an intraabdominal mass (Panel A, arrow) with avidly enhancing solid (Panel B, arrow) and cystic (Panel C, arrow) components and restricted diffusion (Panel D, arrow). Axial and coronal T2-weighted images (Panels E and F, respectively) show normal right and left ovaries (arrows).

euglycemia was maintained during the day with the consumption of frequent meals. She used a continuous glucose monitor and was followed closely by the endocrine service.

PATHOLOGICAL DISCUSSION

Dr. Katelyn C. Dannheim: Core-biopsy specimens showed a spindle-cell proliferation with variable cellularity that was organized in short fascicles with occasional nuclear palisading, admixed with areas of disorganized growth and prominent stro-

mal and perivascular hyalinization (Fig. 3A). The cells were monomorphic with moderate amounts of eosinophilic vacuolated cytoplasm, blunt-ended ovoid nuclei, and finely granular-to-vesicular chromatin with small, inconspicuous nucleoli (Fig. 3B). Mitoses were very rare (<1 mitotic body per 10 high-power fields), and no necrosis was present. On immunohistochemical staining, the lesional cells expressed markers of smooth muscle, including diffuse expression of smooth-muscle actin (Fig. 3C), expression of desmin (Fig. 3D), and dim, variable expression of caldesmon, as well as

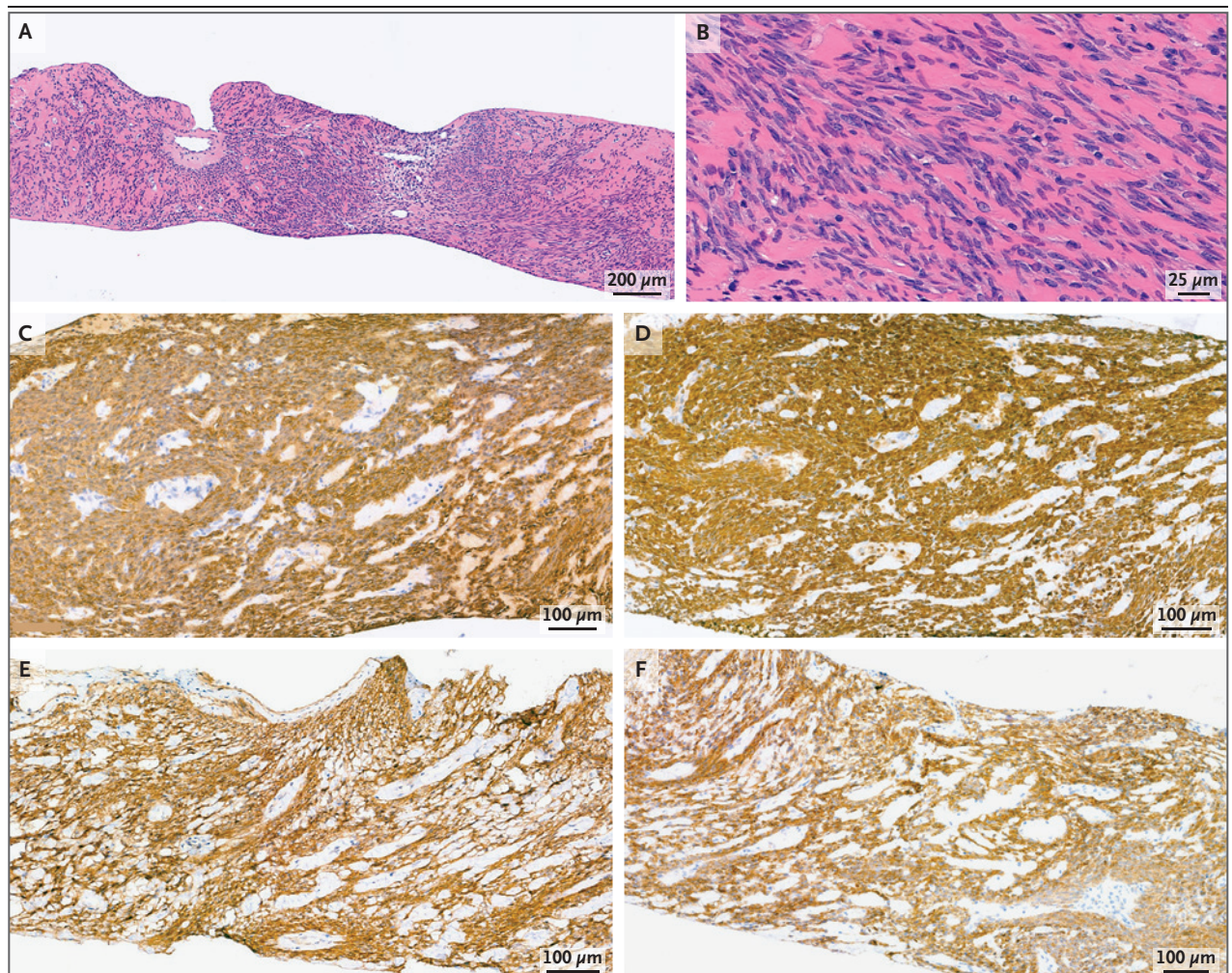


Figure 3. Core-Biopsy Specimens.

Hematoxylin and eosin staining shows a spindle-cell proliferation with variable cellularity that is organized in short fascicles with occasional nuclear palisading, admixed with areas of disorganized growth and prominent stromal and perivascular hyalinization (Panel A). Higher magnification shows that the cells are monomorphic with moderate amounts of eosinophilic vacuolated cytoplasm, blunt-ended ovoid nuclei, and finely granular-to-vesicular chromatin with small, inconspicuous nucleoli (Panel B). Mitoses are very rare (<1 mitotic body per 10 high-power fields), and no necrosis is present. Immunohistochemical staining shows that the lesional cells express markers of smooth muscle, including smooth-muscle actin (Panel C), desmin (Panel D), pan-tropomyosin receptor kinase (pan-TRK; Panel E), and DOG1 (discovered on gastrointestinal stromal tumors protein 1; Panel F).

diffuse expression of pan-tropomyosin receptor kinase (pan-TRK; Fig. 3E) and variable expression of DOG1 (discovered on gastrointestinal stromal tumors protein 1; Fig. 3F). In addition, molecular testing revealed multiple nonspecific sequence variants of potential clinical significance: *ATM* p.G3023D, *CSF3R* p.L48V, *FANCG* p.P319T, *KDR*

p.R1232P, and *RECQL4* p.A717T. No fusions or translocations were detected.

Although the morphologic features and expression of pan-TRK suggested the possibility of a neurotrophic tyrosine receptor kinase (NTRK)-rearranged spindle-cell neoplasm,¹⁴ the absence of S100 and CD34 coexpression and the absence of NTRK fusions ruled out this diagnosis. A gastrointestinal stromal tumor¹⁵ was considered in the context of DOG1 expression, but the strong expression of markers of smooth muscle, the absence of CD117 (c-Kit) expression, the retained expression of succinate dehydrogenase B subunit (SDHB), and the absence of mutations in *KIT*, *PDGFRA*, and the succinate dehydrogenase (SDH) complex did not support this diagnosis. Therefore, a benign smooth-muscle tumor was considered to be the leading diagnosis.

ADDITIONAL DISCUSSION OF SURGICAL MANAGEMENT

Dr. Cameron: Once the results of molecular characterization showed no actionable targets for systemic therapy, the decision was made to proceed with surgical resection of the mass. The patient underwent exploratory laparotomy and was found to have a heterogeneous, well-circumscribed mass that weighed 5 kg emanating from the retroperitoneum. A short segment of sigmoid colon was adherent to the mass and was resected en bloc. There was no invasion of adjacent structures or abnormal lymphadenopathy. The uterus and ovaries appeared normal. A side-to-side functional end-to-end anastomosis was performed in the left colon, and the abdomen was closed.

Intraoperatively, the patient's glucose level normalized immediately after resection of the mass. She was admitted to the surgical intensive care

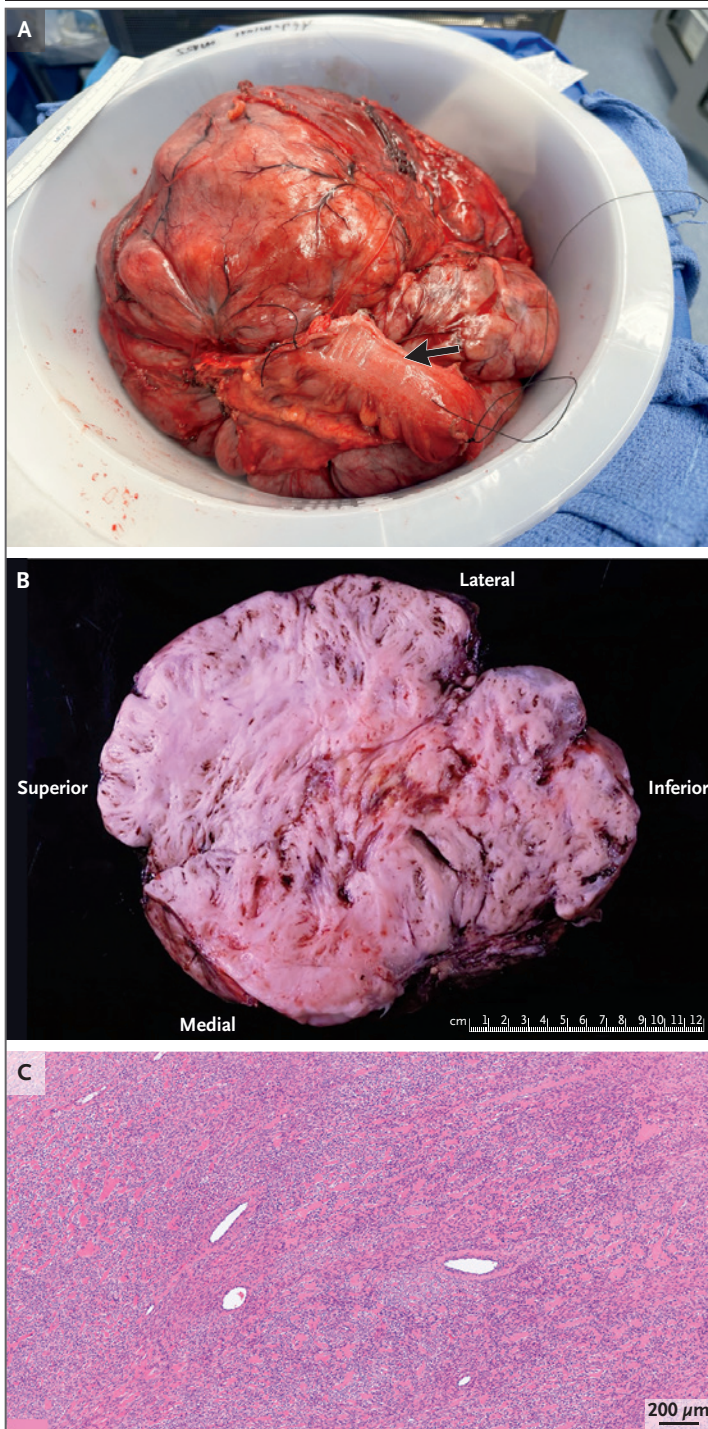


Figure 4. Resection Specimens.

Gross examination of the tumor reveals a lobulated, well-circumscribed mass with smooth surfaces, measuring 28.3 cm by 24.5 cm by 16.5 cm (Panels A and B). The mass was adherent to a segment of sigmoid colon, which was resected en bloc (Panel A, arrow), but further examination showed no evidence of invasion. A cross section from the mass reveals pink-white, rubbery surfaces with a whorled appearance (Panel B). Hematoxylin and eosin staining shows findings that are similar to those of the previous biopsies, with no cytologic atypia, increased proliferation rate, or necrosis that would suggest cancer (Panel C).

unit for frequent glucose monitoring and continued to receive intravenous maintenance fluids while oral intake was initially withheld postoperatively. Her diet was advanced with the return of bowel function, and she had an uncomplicated postoperative course.

ADDITIONAL PATHOLOGICAL DISCUSSION

Dr. Dannheim: Gross (Fig. 4A and 4B) and histologic (Fig. 4C) examinations of the resected specimen were performed. Additional immunohistochemical staining showed calponin expression and retained expression of methylthioadenosine phosphorylase (MTAP), death domain–associated protein 6 (DAXX), adenosine triphosphate–dependent helicase (ATR), phosphatase and tensin homolog (PTEN), and fumarate hydratase (FH). Immunohistochemical staining was negative for mouse double minute 2 homolog (MDM2), retinoblastoma (Rb) protein, and 2-succinocystine (2SC). P53 expression appeared to be lost. On electron microscopy, the tumor cells showed smooth-muscle differentiation. It is notable that DNA methylome and copy-number profiling showed epigenetic features that were similar to those of gastrointestinal stromal tumors and uterine leiomyosarcoma, as well as a complete deletion in the vicinity of the NTRK2 locus, findings that may explain the immunoprofile described earlier. Ultimately, the diagnosis of a cellular smooth-muscle tumor with no histologic features of cancer was made.

PATHOLOGICAL DIAGNOSIS

Benign cellular smooth-muscle tumor.

FOLLOW-UP

Dr. Cameron: On postoperative day 7, the patient was discharged home. She resumed a regular diet, without dextrose supplementation, and euglycemia was restored. She has been followed closely with regular cross-sectional imaging. At the 18-month follow-up, she had no evidence of disease and had a normal glucose level. She had returned to normal activities, including going to school and playing competitive sports.

FINAL DIAGNOSIS

Non-islet-cell tumor hypoglycemia due to a benign cellular smooth-muscle tumor of the abdomen.

This case was presented at Pediatric Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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